# TECHNICAL REPORT

## Predictive Analytics

### Time Series Assignment

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**Abstract**

This assignment studies and analyzes multiple real-world time series with the aim of investigating their characteristics and reliably forecasting future values. Specifically, the following are examined: (a) annual global land temperature anomalies (globtemp), (b) weekly mortality (cmort) in the United States, (c) monthly rainfall in Zevgolatio, Corinthia (2016–2025), (d) quarterly beer sales of an industrial company (2016–2019), and (e) the monthly unemployment rate in Spain (2022–2025). For each time series, diagnostic checks for stationarity, seasonality, and trend are applied, while appropriate models (AR, ARIMA, SARIMA) are fitted, evaluated on the basis of AIC/BIC criteria, and used to produce forecasts with confidence intervals. The results highlighted the importance of proper preliminary analysis for model selection and forecast reliability, while also revealing limitations of classical statistical models when dealing with highly volatile data or extreme values.

# Contents

# 1 Introduction

Time series analysis is one of the most important tools in statistics and data science, with broad applications in economics, meteorology, health, and technology [5]. In this assignment, real-world time series datasets are examined with the aim of understanding their dynamics, detecting trends and seasonality, and forecasting future values through suitable stochastic models such as ARIMA and SARIMA. Preliminary analysis techniques, stationarity checks, and diagnostic tests for the selected models are presented, with particular emphasis on evaluating their forecasting ability.

The structure of the assignment begins with a brief theoretical overview of the main concepts, followed by a detailed description of the data and methodology. The appropriate statistical analysis is then applied separately to each time series, with a detailed presentation of the results and conclusions.

# 2 Literature Review

Time series analysis is a fundamental tool for understanding and forecasting phenomena that evolve over time [10]. A time series is a sequence of measurements recorded at successive points in time and often exhibits dependence between consecutive values [5].

**Stationarity:** An important prerequisite in time series analysis is stationarity, meaning that statistical characteristics such as the mean and variance remain stable over time [10]. Lack of stationarity, for example due to trend or seasonality, is often addressed through suitable transformations such as differencing.

**Seasonality:** Many time series exhibit seasonality, that is, periodic patterns that repeat at a specific interval (e.g., annually or monthly) [3]. Seasonality analysis helps identify and remove these periodic effects in order to improve forecasting.

**ARIMA/SARIMA Models:** ARIMA (AutoRegressive Integrated Moving Average) models are widely used for time series forecasting because they can handle both stationary and non-stationary series [1]. SARIMA (Seasonal ARIMA) models extend ARIMA by incorporating seasonal terms so that the seasonality of the series can also be modeled [11].

The selection and correct application of these models require appropriate diagnostic tests and the use of statistical criteria such as AIC and BIC in order to achieve the most suitable fit to the data [5].

# 3 Data and Methodology

## 3.1 Data

This assignment studies and analyzes multiple real-world time series with the aim of investigating their characteristics and reliably forecasting future values. Specifically, the following datasets are examined:

- **Annual global land temperature anomalies (globtemp):** Annual data from 1850 to 2023 recording changes in average global land temperature.
- **Weekly mortality (cmort) in the United States:** Weekly mortality observations that are often used to analyze epidemiological phenomena and forecasting methods.

- **Monthly rainfall in Zevgolatio, Corinthia:** Monthly rainfall values for the years 2016–2025, collected from meteorological data for the region.
- **Quarterly beer sales of an industrial company:** Sales data, in millions of bottles, by season (Winter, Spring, Summer, Autumn) for the years 2016–2019.
- **Monthly unemployment rate in Spain:** Monthly unemployment data from Eurostat and INE for the period January 2022 to April 2025.

## 3.2 Methodology

The time series analysis includes the following steps:

- **Preliminary analysis**: Visualization and statistical examination of stationary or non-stationary behavior using plots, ACF/PACF, and stationarity tests.
- **Decomposition and preprocessing**: Removal of trend and seasonality where necessary through differencing or decomposition (multiplicative/additive decomposition).
- **Application of suitable models**: Fitting AR, MA, ARMA, ARIMA, or SARIMA models according to the characteristics of each series.
- **Diagnostic checking**: Testing residuals for white-noise behavior (Ljung-Box test), evaluating error distributions (Q-Q plots and histograms), and calculating evaluation metrics (MSE, RMSE, MAE, AIC, BIC).
- **Forecasting**: Estimating future values with confidence intervals and visualizing the results.
- **Comparison and conclusions**: Summarizing results, comparing models, and interpreting the findings for each time series.

# 4 Exercise 1: Time Series Analysis of Unemployment in Spain

## 4.1 Data Description

For the analysis of the unemployment-rate time series, monthly unemployment data for Spain covering the period from January 2022 to April 2025 were used. The data were collected from official statistical sources such as Eurostat [4] and INE – Instituto Nacional de Estadística [6], providing a total of 40 consecutive monthly observations. Each observation corresponds to the unemployment rate (%) of Spain's labor force for the respective month.

This period was selected in order to include possible short-term fluctuations, longer-term trends, and the effects of seasonal factors such as the tourism season and winter slowdown. The data form a one-dimensional numerical sequence (time series), where each element corresponds to one month and has the structure: [Year, Month, Unemployment Rate (%)].

Table 1 presents examples of the first and last values of the series, while the following plot (Figure 1) shows the evolution of Spain's unemployment rate over the period under study.

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**Figure 1:** Monthly evolution of the unemployment rate in Spain (Jan. 2022 – Apr. 2025)

Sample data format:

**Table 1:** Sample Monthly Unemployment Rate Values in Spain (2022–2025)

| **Year** | **Month** | **Unemployment Rate (%)** |
|---|---|---|
| 2022 | 01 | 13.6 |
| 2022 | 02 | 13.8 |
| 2022 | 03 | 13.8 |
| 2022 | 04 | 13.1 |
| 2022 | 05 | 12.5 |
| 2022 | 06 | 12.4 |
| ⋮ | ⋮ | ⋮ |
| 2025 | 01 | 11.2 |
| 2025 | 02 | 11.5 |
| 2025 | 03 | 11.5 |
| 2025 | 04 | 10.9 |

This time series exhibits pronounced fluctuations, annual seasonality, and a clear downward trend, as discussed in the following sections.

## 4.2 Time Plot & Seasonality

The evolution of Spain's unemployment rate from January 2022 to April 2025 shows a strong downward trend, with unemployment decreasing from approximately 13.8% at the beginning of 2022 to 10.9% in April 2025, as shown in Figure 2.

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**Figure 2:** Monthly evolution of the unemployment rate in Spain (Jan. 2022 – Apr. 2025)

This time series is characterized as non-stationary because both its mean and variance change substantially over time, while strong seasonality is also observed. The seasonality analysis and corresponding plots (Figure 3) reveal a recurring annual pattern, with clear unemployment peaks from January to March and lower levels during May–June.

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**Figure 3:** Time series decomposition: trend, seasonality, and random component

This seasonality is confirmed both by the monthly boxplots (Figure 4), which show notable differences between monthly values, and by the autocorrelation function (ACF, Figure 5), which exhibits peaks at 12-month intervals.

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**Figure 4:** Boxplot of the seasonal variation in the unemployment rate by month

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**Figure 5:** Autocorrelation function (ACF) of the unemployment time series

The observed fluctuations are mainly associated with the seasonal nature of the Spanish economy, as the tourism season increases employment during the summer months, whereas holiday periods and winter are associated with temporary layoffs and higher unemployment.

In conclusion, the analysis of Spain's unemployment time series reveals both a substantial downward trend and strong annual seasonality, indicating the need for employment policies that take these structural and cyclical factors into account.

## 4.3 Stationarity Testing & ARMA Fitting

To investigate the stationarity of Spain's unemployment-rate time series, the autocorrelation (ACF) and partial autocorrelation (PACF) plots were examined. The original ACF and PACF plots showed significant values at many consecutive lags, indicating that the time series is not stationary. First differences were therefore applied to obtain a stationary series. The plot of the first differences (Figure 8) shows that the observations are distributed around zero, a characteristic of a stationary series.

The stationarity of the transformed series is further supported by the ACF of the first differences (Figure 6), where the values decrease rapidly after lag 1, and by the PACF (Figure 7), where one significant value is observed at the first lag. This pattern suggests that the first-differenced series can be appropriately modeled with an ARMA model. ARMA(2,1) and ARMA(1,1) models were tested, using the Akaike Information Criterion (AIC) and residual analysis as selection criteria. The ARMA(2,1) model produced a lower AIC, all coefficients were statistically significant, and its residuals showed white-noise characteristics, confirming a satisfactory fit (Figure 9).

The forecast of the first differences using the selected model for the following months is shown in Figure 10. The predicted values remain close to zero, which is consistent with the stationary nature of the differenced series. Overall, the stationarity procedure and the ARMA(2,1) fit provide a reliable basis for short-term forecasting of the unemployment time series.

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**Figure 6:** ACF of the first differences of the unemployment time series

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**Figure 7:** PACF of the first differences of the unemployment time series

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**Figure 8:** Plot of the first differences of the unemployment rate (stationary series)

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**Figure 9:** ACF and PACF of the residuals of the fitted ARMA model

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**Figure 10:** Forecast of the first differences of the unemployment rate using the ARMA(2,1) model

## 4.4 First Differences and Stationarity Testing of the New Series

Following the finding that the original unemployment-rate time series is non-stationary, the first-difference series was calculated, that is, the difference between each value and the previous one. This series is shown in Figure 8. The autocorrelation (ACF) and partial autocorrelation (PACF) functions were then re-examined for the transformed series (Figures 6, 7). The results show that both the ACF and PACF decay rapidly, indicating that the first-difference series is now stationary. It is therefore suitable for ARMA or ARIMA modeling.

## 4.5 Fitting and Evaluation of an ARIMA(2,1,1) Model

An ARIMA(2,1,1) model was fitted to the monthly unemployment data for Spain over the analysis period. The estimated model parameters are summarized in Table 2.

**Table 2:** Parameter Estimates of the ARIMA(2,1,1) Model

| **Parameter** | **Estimate** | **Std. Error** | **z-value** | **p-value** |
|---|---|---|---|---|
| ar1 | 1.02 | 0.18 | 5.64 | $< 0.001$ |
| ar2 | -0.68 | 0.13 | -5.29 | $< 0.001$ |
| ma1 | -0.32 | 0.22 | -1.46 | 0.143 |

**Discussion:** The ar1 and ar2 coefficients are statistically significant ($p$-value $< 0.05$), whereas the ma1 coefficient is not statistically significant, but it is retained in the model to improve the fit.

**Residual Diagnostics:** Diagnostic examination of the residuals through autocorrelation (ACF) and partial autocorrelation (PACF) plots did not reveal strong autocorrelation, while the Box-Ljung test ($p$-value $= 0.024$) indicates borderline statistical significance. This suggests that some weak residual correlation may remain and is not fully captured by the model.

**Forecast and Plot:** Figure 11 presents the historical unemployment-rate time series (black line), the model forecasts for the following 12 months (red line), and the 95% confidence intervals (blue dashed lines).

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**Figure 11:** Forecast of unemployment in Spain using the ARIMA(2,1,1) model.

**Discussion:** The forecast indicates relative stabilization of the unemployment rate over the following year, without extreme fluctuations. The confidence intervals widen as the forecast horizon increases, as expected because uncertainty grows over longer forecasting horizons.

**Conclusion:** The ARIMA(2,1,1) model provides a satisfactory fit to the data and reliable short-term forecasts for unemployment in Spain. There are minor indications of residual autocorrelation, which could potentially be addressed by testing more complex models or incorporating additional exogenous variables if available.

## 4.6 Testing for Second Differences

According to the results of Sections 4.4 and 4.5, the first-difference series of the original time series was found to be already stationary, as indicated by the ACF and PACF analyses. Therefore, second differences are not required, nor is further stationarity testing of the transformed series necessary. The stationary first-difference series is considered suitable for ARMA or ARIMA modeling, as applied in the previous sections.

## 4.7 Final ARIMA Model Fitting to the Original Series

As established from the analysis of the first differences of the unemployment-rate time series, the transformed series is clearly stationary and is therefore appropriate for ARMA or ARIMA models with $d = 1$. Within this framework, an **ARIMA(2,1,1)** model was fitted to the original data, incorporating one difference ($d = 1$). The model results indicate a satisfactory fit and adequate attenuation of residual autocorrelation, while the resulting forecasts exhibit realistic behavior and the expected uncertainty over the forecast horizon. Consequently, the ARIMA(2,1,1) model is considered the most appropriate choice for Spain's unemployment time series over the period examined.

## 4.8 Model Adequacy Check and Comparison with Extended Models

To evaluate the adequacy of the final **ARIMA(2,1,1)** model, residual analysis was performed and the model was compared with higher-order alternatives.

**(I) Residual Check:** The autocorrelation (ACF) and partial autocorrelation (PACF) plots of the model residuals (Figure 12) show that the values remain within the significance limits, with no evidence of strong autocorrelation. However, the Box-Ljung test produces a p-value slightly below 0.05 ($p = 0.024$), indicating the possible presence of mild residual structure that is not fully captured by the model.

**(II) Comparison with Extended Models:** Additional higher-order ARIMA models (ARIMA(3,1,1), ARIMA(2,1,2), ARIMA(2,2,1)) were fitted and compared using the AIC criterion and the Box-Ljung p-value for the residuals. The results are summarized in Table 3.

**Table 3:** Comparison of ARIMA models in terms of AIC and Box-Ljung p-value.

| Model | AIC | Box-Ljung p-value |
|---|---|---|
| ARIMA(2,1,1) | 13.48 | 0.024 |
| ARIMA(3,1,1) | 12.05 | 0.071 |
| ARIMA(2,1,2) | **5.88** | 0.0003 |
| ARIMA(2,2,1) | 17.79 | 0.0073 |

The **ARIMA(2,1,2)** model has the lowest AIC, but the Box-Ljung p-value is extremely small (0.0003), indicating significant autocorrelation in the residuals and therefore an inadequate model. In contrast, the **ARIMA(3,1,1)** model has a relatively low AIC and an acceptable p-value (0.071), indicating better statistical adequacy than the ARIMA(2,1,1) model.

**Conclusion:** Although the initial ARIMA(2,1,1) model captures the main dynamics of the time series, the extended ARIMA(3,1,1) model provides slightly better residual adequacy. The final model should therefore be selected on the basis of a combination of adequacy measures (AIC, Box-Ljung test) and interpretability, with preference given to ARIMA(3,1,1) when statistical adequacy is the main priority. A low AIC indicates good fit, but the absence of autocorrelation in the residuals is essential for the validity of the model forecasts. Thus, when there is a conflict between AIC and statistical adequacy, the model with better residual behavior is preferred. In statistical time series modeling, final model selection is based not only on fit indices such as AIC, but also on the absence of structure in the residuals so that the assumptions of the model are satisfied.

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**Figure 12:** ACF and PACF of the residuals for the ARIMA(2,1,2) model.

## 4.9 Long-Term Unemployment Forecast

To forecast the future path of Spain's unemployment rate, the final **ARIMA(3,1,1)** model was applied. This model was selected as the most appropriate among several candidate models on the basis of statistical criteria (low AIC and satisfactory residual adequacy).

Figure Figure 13 presents the unemployment-rate forecast for the next 24 months:

- **The black line** represents the historical observed data.
- **The red line** represents the predicted unemployment-rate values based on the selected ARIMA(3,1,1) model.
- **The blue dashed lines** define the 95% confidence interval of the forecasts.

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**Figure 13:** Forecast of the unemployment rate for the next 24 months using the ARIMA(3,1,1) model.

**Discussion of Results:**

- **Relative stabilization:** The unemployment-rate forecast remains at levels similar to the most recent observations, without showing a strong upward or downward trend.

- **Mild fluctuation:** Small cyclical fluctuations can be observed in the predicted values, possibly reflecting seasonal patterns inherited from the historical sample.
- **Uncertainty:** The 95% confidence intervals widen as the forecast horizon increases, reflecting the normal increase in uncertainty over longer forecasting horizons.
- **Absence of extreme values:** No extreme forecast values or exponential upward/downward trends are observed, suggesting that the model produces conservative and realistic forecasts.

**Conclusion:** The final **ARIMA(3,1,1)** model provides reliable short- and medium-term forecasts for the unemployment rate, showing relative stability and satisfactory interpretability. Forecast accuracy decreases, as expected, as the forecast horizon increases, which is reflected by the widening confidence intervals.

# 5 Exercise 2: Analysis of Climate Data

## 5.1 Data Description

For the analysis of monthly rainfall in Zevgolatio, Corinthia, during the period 2016–2025, open meteorological data obtained from the open-meteo.com platform [9] were used. The original data consisted of daily rainfall values (in millimeters, mm) for each calendar day of the study period.

Initially, the data were imported into the R environment and cleaned so that metadata rows were removed during file reading. The date column (time) was converted to the Date type, and the year and month were then extracted from each record. To create the monthly time series, the daily values were grouped by year and month and the total rainfall for each month was calculated using the aggregate function in R.

The final result is a one-dimensional time series representing monthly rainfall depth (in mm) for each month from January 2016 to June 2025. The following plot (Figure 14) illustrates the evolution of rainfall over the study period.

These data form the basis for the time series analysis and rainfall forecasting for the region.

## 5.2 Time Plot, Stationarity and Seasonality

Figure 14 presents the time plot of monthly rainfall in Zevgolatio, Corinthia, for the period 2016–2025.

From the plot, the following observations can be made:

- There are large variations in rainfall from year to year, as well as some periods with particularly heavy rainfall. This suggests that the mean and variance of the series change over time; therefore, the time series cannot be considered stationary.
- Repeating within-year patterns are also visible, with higher rainfall during the winter months and lower rainfall during summer. This provides clear evidence of annual seasonality in the series.

In conclusion, monthly rainfall exhibits strong seasonality and non-stationarity, because the statistical properties of the series (mean and variance) vary over time and are influenced by the seasonal climatic conditions of the region.

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**Figure 14:** Monthly rainfall (in mm) in Zevgolatio, Corinthia, for the period 2016–2025, as calculated from the daily data obtained from open-meteo.com.

## 5.3 Stationarity Examination Using ACF and PACF

To investigate the stationarity of the monthly rainfall time series for Zevgolatio, Corinthia, the autocorrelation (ACF) and partial autocorrelation (PACF) plots were examined, as shown in Figures 15 and 16.

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**Figure 15:** Autocorrelation function (ACF) plot of monthly rainfall in Zevgolatio, Corinthia.

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**Figure 16:** Partial autocorrelation function (PACF) plot of monthly rainfall in Zevgolatio, Corinthia.

**Discussion:** The ACF shows a very high value at the first lag, after which the values decrease relatively slowly, with most subsequent lags remaining within the confidence intervals. Similarly, the PACF shows a significant value only at the first lag, while the remaining lags are low.

This pattern, together with the time plot of the series, indicates that the time series is not stationary. The significant autocorrelation at lag 1 and the slow decay of the values are indications of non-stationarity, while the time plot in the previous section also revealed strong seasonality and changes in the mean.

**Conclusion:** The monthly rainfall time series for Zevgolatio, Corinthia, is not stationary. To achieve stationarity, an appropriate transformation such as first-order differencing and/or seasonal differencing should be applied. Therefore, in its current form the series is not suitable for the direct application of an ARMA$(p, q)$ model. It must first be transformed into a stationary series, after which the autocorrelation plots should be re-examined to select an appropriate model.

## 5.4 Stationarity Test of First Seasonal Differences

To investigate stationarity, first seasonal differences (lag=12) were applied to the monthly rainfall series for Zevgolatio, Corinthia. The resulting series was analyzed using both statistical criteria (mean and variance) and time series, ACF, and PACF plots.

**Statistical Criteria:**

- **Annual Means (mm):**
  1.61, -14.35, -6.29, -9.35, -10.08, 15.97, 13.38, 3.76, 14.83
  The mean values vary substantially, alternating between positive and negative values, which indicates that the mean of the series does not remain stable over time.

- **Annual Variances (mm$^2$):**
  1767.82, 1357.72, 1526, 1070.89, 421.42, 3375.7, 3682.91, 2166.06, 2140.49
  The variance differs considerably from year to year, ranging from 421.42 to 3682.91, which further supports the hypothesis of non-stationarity. The criterion $\max(\text{vars})/\min(\text{vars}) = 3682.91/421.42 \approx 8.74 > 4$ also indicates non-constant variance.

**Graphical Analysis:**

- **Time Plot:** Figure 17 presents the first seasonal differences, which exhibit strong fluctuations, large annual peaks, and substantial changes in the mean.

- **ACF and PACF:** The ACF and PACF plots (Figure 18) show a slow decay of the values, as well as significant values at several lags, particularly at seasonal lags such as lag=12. This is characteristic of a non-stationary series because, in a stationary series, the ACF/PACF values decay rapidly.

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**Figure 17:** Time plot of the first seasonal differences (lag=12) of monthly rainfall.

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**Figure 18:** ACF and PACF of the first seasonal differences of monthly rainfall.

**Conclusion:** The first seasonal difference series (lag=12) is **not stationary**, because both the mean and variance vary considerably across years, while the ACF and PACF show significant values at high lags. To achieve stationarity, further transformation is required, for example additional differencing or another suitable transformation.

## 5.5 Suitability for SARIMA Modeling

According to the analysis in the previous section, the first seasonal difference series is not stationary, because both the mean and variance vary substantially over time and the ACF and PACF plots show slow decay and significant values at high lags. Stationarity is a fundamental requirement for the correct application of ARMA/ARIMA/SARIMA time series models.

Therefore, a SARIMA model cannot be fitted directly to the original series, since the stationarity assumption is not satisfied even after applying first seasonal differences.

To make SARIMA modeling feasible, further transformations should be tested, such as second differencing, or the possible influence of exogenous factors on the structure of the series should be investigated.

**Conclusion:** The continued non-stationarity of the series after seasonal differencing indicates a complex structure and requires further statistical investigation before applying SARIMA models.

## 5.6 Stationarity Test of Double Differences (First + Seasonal) for Monthly Rainfall

According to the previous question, the first seasonal difference series was not considered stationary. For this reason, **double differencing** was applied to the series: first differences together with seasonal differences (lag=12). The resulting time series is analyzed below with respect to stationarity using both graphical examination and the autocorrelation (ACF) and partial autocorrelation (PACF) plots.

**Time Plot:** Figure 19 shows the double-differenced series. The new time series fluctuates around zero without an evident trend or strong periodic fluctuations. The mean and variance appear to remain stable over time, suggesting possible stationarity.

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**Figure 19:** Time plot of the double differences (first + seasonal) of monthly rainfall in Zevgolatio, Corinthia.

**ACF and PACF Analysis:** The autocorrelation (ACF, Figure 20) and partial autocorrelation (PACF, Figure 21) plots show the following:

- **ACF:** A significant negative value appears at lag 1 ($-0.45$), along with a noticeable value at lag 12 ($-0.35$), while the remaining lags are close to zero. The rapid decay of the coefficients indicates the absence of long-term autocorrelation, which is a characteristic of a stationary series.

- **PACF:** The PACF shows significant values at lags 1 and 2, as well as a negative value at lag 12. The rapid reduction in the values also supports stationarity.

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**Figure 20:** ACF of the double differences (first + seasonal) of monthly rainfall.

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**Figure 21:** PACF of the double differences (first + seasonal) of monthly rainfall.

**Conclusion:** The double-differenced series (first + seasonal) of monthly rainfall exhibits the characteristics of a **stationary time series**, because it shows neither trend nor seasonality, while the ACF/PACF values decay rapidly and there are no strong autocorrelations at high lags.

## 5.7 Fitting and Evaluation of the SARIMA Model

Based on the preceding stationarity findings, the SARIMA$(2,1,1)(1,1,1)_{12}$ model was fitted to the monthly rainfall series for Zevgolatio, Corinthia, for the years 2016–2025. The model includes first differences ($d = 1$), first seasonal differences ($D = 1$), and a seasonal period of 12 months.

**Estimated Model Parameters:**

- AR(1): $-0.003$
- AR(2): $-0.085$
- MA(1): $-0.877$
- SAR(1): 0.136
- SMA(1): $-1.00$

**Model Statistics:**

- $AIC$: 1030.77
- log-likelihood: $-509.39$

**Residual Diagnostic Analysis:**

To evaluate the adequacy of the SARIMA model, the residuals were analyzed with respect to stationarity, autocorrelation, and normality.

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**Figure 22:** Autocorrelation (ACF, left) and partial autocorrelation (PACF, right) plots of the SARIMA model residuals.

The ACF and PACF plots of the residuals show that all values lie within the confidence intervals, there is no significant autocorrelation, and the residuals can be regarded as white noise.

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**Figure 23:** Histogram of the SARIMA model residuals. The distribution is approximately normal, with slight asymmetry.

**Ljung-Box Test:** For the model residuals, the Ljung-Box test at lag=24 gives:

$X^2 = 17.22$, $p$-value $= 0.839$

Because the $p$-value is much greater than 0.05, **the null hypothesis of residual independence is not rejected**. The model residuals do not exhibit statistically significant autocorrelation.

**Monthly Rainfall Forecast:**

Using the estimated SARIMA model, monthly rainfall was forecast for the next 12 months. Figure 24 presents the observations, predicted values, and 95% confidence intervals.

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**Figure 24:** Forecast of monthly rainfall for the next 12 months using the SARIMA model. The black line represents the actual observations, the red line the forecasts, and the blue dashed lines the 95% confidence interval.

**Discussion:** The SARIMA$(2,1,1)(1,1,1)_{12}$ model was successfully fitted to the data and provides a satisfactory representation of the dynamics of the series. The forecast displays seasonal patterns and realistic uncertainty intervals. The confidence intervals increase further into the future, reflecting the expected growth of uncertainty in longer-term forecasts.

**Conclusion:** The SARIMA$(2,1,1)(1,1,1)_{12}$ model provides a reliable forecast of monthly rainfall, with satisfactory residual behavior and consistent statistical results.

## 5.8 Application and Comparison of Extended Models

The adequacy of the final SARIMA$(2,1,1)(1,1,1)_{12}$ model for monthly rainfall was systematically examined in two ways: (I) detailed residual diagnostics and (II) comparison with higher-order extended models.

### (I) Residual Diagnostic Analysis

- **ACF/PACF Plots:** The autocorrelation (ACF) and partial autocorrelation (PACF) plots of the model residuals (Figure 22) show that all values are within the confidence limits, with no significant peaks at seasonal lags. This indicates that the residuals approximate white noise.
- **Q-Q Plot:** The Q-Q plot (Figure 25) follows the reference line reasonably well, with small deviations at the tails, which is considered acceptable for hydrological data.
- **Ljung-Box Test:** The Ljung-Box test at lag=24 produced a $p$-value of 0.839 (i.e., $> 0.05$), so the white-noise hypothesis for the residuals is not rejected.

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**Figure 25:** Q-Q plot of the SARIMA model residuals. The distribution is close to normal, without strong asymmetry or heavy tails.

### (II) Comparison with Extended Models

To determine whether a more complex model would perform better, the following extended models were also fitted:

- SARIMA$(3,1,1)(1,1,1)_{12}$
- SARIMA$(2,1,2)(1,1,1)_{12}$
- SARIMA$(2,1,1)(2,1,1)_{12}$
- SARIMA$(2,1,1)(1,1,2)_{12}$

- SARIMA$(3,1,2)(2,1,2)_{12}$

The results (AIC and Ljung-Box test) are summarized in Table 4.

**Table 4:** Comparison of adequacy measures between the baseline and extended SARIMA models

| **Model Specification** | **AIC** | $\Delta$AIC | **Ljung-Box p-value** |
|---|---|---|---|
| SARIMA(2,1,1)(1,1,1)$_{12}$ | 1030.77 | 0.000 | 0.839 |
| SARIMA(2,1,1)(1,1,2)$_{12}$ | 1031.03 | +0.26 | 0.742 |
| SARIMA(3,1,1)(1,1,1)$_{12}$ | 1031.26 | +0.48 | 0.681 |
| SARIMA(3,1,2)(2,1,2)$_{12}$ | 1032.07 | +1.30 | 0.598 |
| SARIMA(2,1,1)(2,1,1)$_{12}$ | 1032.62 | +1.85 | 0.553 |
| SARIMA(2,1,2)(1,1,1)$_{12}$ | 1032.99 | +2.22 | 0.512 |

**Conclusions:**

- The original SARIMA$(2,1,1)(1,1,1)_{12}$ model had the lowest AIC and the most white-noise-like residuals (highest Ljung-Box p-value).
- None of the extended models produced a substantial improvement (all had $\Delta$AIC $< 2$).
- The diagnostic checks indicate that adding extra parameters leads to overfitting without meaningful benefit.

**Final Assessment:**
The SARIMA$(2,1,1)(1,1,1)_{12}$ model is considered fully adequate and suitable for operational forecasting of monthly rainfall. It maintains parametric simplicity and statistical reliability while effectively capturing the dynamics of the rainfall time series.

## 5.9 Monthly Rainfall Forecast for the Next Two Years

Using the final SARIMA$(2,1,1)(1,1,1)_{12}$ model, monthly rainfall was forecast for the following two years (2025–2027). The results are summarized in the plots and tables below.

The figure is not available.

**Figure 26:** Forecast of monthly rainfall for the next two years using the SARIMA model. The black line represents historical data, the red line the forecast, the blue dashed lines the 95% confidence intervals, and the green vertical line the start of the forecast period.

**Forecast Evaluation:**

- **Seasonality and variability:** The forecasts follow the seasonal patterns of historical rainfall, with fluctuations similar to those observed in previous years.
- **Confidence intervals:** The confidence intervals widen further into the future, reflecting increasing forecast uncertainty.
- **Comparison with historical data:** The forecast values show a higher average rainfall level compared with the historical mean.

The figure is not available.

**Figure 27:** Comparison of the forecast annual mean rainfall with the historical mean.

| Year | Forecast Mean (mm) | Difference from Historical Mean |
|---|---|---|
| First Year | — | +56.6% |
| Second Year | — | +60.7% |

**Table 5:** Percentage difference between the annual forecast mean and the historical mean for the two forecast years.

The figure is not available.

**Figure 28:** Comparison of actual (black line) and predicted (red line) rainfall values over the evaluation period.

The figure is not available.

**Figure 29:** Comparison of monthly forecasts by year with the historical mean (blue line).

**Comparison of Actual and Predicted Values:**

The figure is not available.

**Figure 30:** Autocorrelation function (ACF) of the forecast errors. All values lie within the confidence intervals, with no significant autocorrelation.

**Forecast Diagnostic Checks:**

**Conclusions:**

- The model forecasts preserve the dynamics and seasonality of the time series, producing realistic and stable ranges of values for the next two years.
- The mean forecast rainfall is substantially higher, by approximately 57–61%, than the historical mean, suggesting a possible change in climatic behavior or the influence of exogenous factors.
- The confidence intervals become wide over longer horizons, making continuous updating of the model with new data important.

**Final Comment:** The long-term forecast is useful as an indication of potential trends; however, uncertainty remains high because of the nature of rainfall time series and the inherent variability of climate.

# 6 Exercise 3: Analysis of cmort, globtemp, and Beer Sales

## 6.1 Description and Modeling of the Weekly Mortality Time Series `cmort`

### 6.1.1 Description of the `cmort` Data

The `cmort` time series refers to weekly mortality (total deaths per week) in the United States and consists of weekly observations covering the period 1970–1979. These data are often used as an example in time series analysis for learning forecasting, seasonality, and autocorrelation models.

The `cmort` dataset contains 508 consecutive weekly values recording the number of deaths from all causes in the Los Angeles area. The series exhibits seasonality, fluctuations related to infectious diseases such as influenza, and possibly a longer-term trend.

The data are available in the widely used `astsa` package in R and are described extensively in the time series literature [10].

> *The `cmort` data are widely used in textbooks and university courses as a standard example for time series analysis with autoregressive and moving-average models.*

### 6.1.2 Fitting an AR(2) Model to the `cmort` Time Series and Short-Term Forecasting

A second-order autoregressive model, AR(2), was fitted to the weekly mortality time series `cmort` using linear regression. The general form of the model is:

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \epsilon_t$$

where $\epsilon_t$ is white noise.

**Model Estimation:** Using the least-squares method, the fitted AR(2) model has the form:

$$\widehat{x}_t = \widehat{\phi}_1 x_{t-1} + \widehat{\phi}_2 x_{t-2}$$

and the estimated residuals are defined as $\widehat{\epsilon}_t = x_t - \widehat{x}_t$.

**Diagnostic Checks:**

- **Residual ACF/PACF:** The autocorrelation (ACF) and partial autocorrelation (PACF) plots of the residuals (Figure 31) show that all values lie within the confidence limits, indicating that the residuals behave as white noise.
- **Ljung-Box Test:** For $lag = 10$ and $lag = 20$, the p-values are 0.6141 and 0.3902, respectively. Therefore, the null hypothesis of independence (white noise) is not rejected.
- **Visual Examination of Residuals:** The residuals have approximately zero mean, stable variance, and no visible structure (Figure 32).

The figure is not available.

**Figure 31:** ACF (left) and PACF (right) plots of the AR(2) model residuals.

The figure is not available.

**Figure 32:** Time plot of the AR(2) model residuals.

**Model Accuracy Measures:**

- Mean Squared Error (MSE): 32.32
- Root Mean Squared Error (RMSE): 5.68
- Mean Absolute Error (MAE): 4.48

**Short-Term Forecast and 95% Confidence Interval:** Using the estimated model, mortality was forecast for the next four weeks. Figure 33 displays the predicted values and the 95% confidence intervals:

The figure is not available.

**Figure 33:** Mortality forecast for four weeks using AR(2) with 95% confidence intervals (green lines).

**Conclusions:**

- The AR(2) model describes the structure of the `cmort` time series satisfactorily, with residuals showing white-noise characteristics.
- The forecast for the next four weeks is considered reliable within the given confidence interval.
- The confidence intervals are relatively wide, reflecting the natural uncertainty associated with mortality forecasting.
- For more extreme fluctuations or unusual mortality episodes, the simple AR(2) model may not be sufficient.

## 6.2 Analysis and Forecasting of the Global Land Temperature Time Series Using ARIMA

### 6.2.1 Description of the Global Land Temperature Time Series (`globtemp`)

The `globtemp` time series contains annual measurements of global land temperature anomalies for the period 1850–2023. The data originate from international climate databases and represent one of the most important indicators for studying long-term trends in the Earth's land temperature in relation to climate change. More specifically, `globtemp` records the difference between the mean temperature of each year and a predefined reference mean (climatological baseline), thereby enabling comparisons across periods and the identification of long-term changes in temperature [2, 7].

Global temperature time series are widely used in statistics and climate econometrics to analyze trends, anomalies, and future values [8]. The analysis of such series requires stationarity testing, trend removal, and the selection of suitable stochastic models such as ARIMA to ensure valid statistical conclusions.

### 6.2.2 Preliminary Analysis and Stationarity

The `globtemp` time series, which contains annual global land temperature data for the years 1850–2023, was first examined. A clear upward temperature trend was observed. This was also supported by the autocorrelation (ACF) and partial autocorrelation (PACF) plots of the original series, where the values decay slowly, indicating non-stationarity (Figure 34).

The figure is not available.

**Figure 34:** ACF and PACF of the original globtemp time series.

To achieve stationarity, the first difference of the series was calculated, as shown in Figure 35. The corresponding ACF and PACF plots (Figure 36) indicate that the differenced series now exhibits stationary characteristics.

The figure is not available.

**Figure 35:** First difference of the globtemp time series.

The figure is not available.

**Figure 36:** ACF and PACF of the first difference of the globtemp series.

#### 6.2.3 Selection of a Suitable ARIMA Model

Several ARIMA models with one difference ($d = 1$) were tested. Selection was based on the Akaike Information Criterion (AIC):

| Model | d | Coefficients | AIC |
|---|---|---|---|
| ARIMA(1,1,0) | 1 | ar1 = -0.065 | 152.72 |
| ARIMA(1,1,1) | 1 | ar1 = -0.066, ma1 = -0.73 | 120.64 |
| ARIMA(2,1,1) | 1 | ar1, ar2, ma1 | 122.58 |

**Table 6:** Comparison of ARIMA models and AIC values.

The best model is ARIMA(1,1,1), which has the lowest AIC.

### 6.2.4 Model Diagnostic Analysis

The diagnostics of the ARIMA(1,1,1) model include residual analysis for normality and autocorrelation:

- **Residual Histogram:** The residuals are approximately symmetric and normally distributed (Figure 37).
- **Q-Q Plot:** The points approximately align with the normal reference line (Figure 38).
- **Residual ACF:** No significant autocorrelation is observed.
- **Ljung-Box Test:** The $p$-value is $> 0.05$, indicating that the residuals behave as white noise.

The figure is not available.

**Figure 37:** Histogram of the residuals of the ARIMA(1,1,1) model.

The figure is not available.

**Figure 38:** Q-Q plot of the ARIMA(1,1,1) residuals.

The figure is not available.

**Figure 39:** Residuals of the ARIMA(1,1,1) model by year.

### 6.2.5 Forecast for the Next 10 Years

Using the final ARIMA(1,1,1) model, forecasts were produced for the years 2024–2033. The 95% confidence interval was calculated and visualized in Figure 40.

The figure is not available.

**Figure 40:** Forecast of global land temperature using ARIMA(1,1,1) for the next 10 years (red line: forecast, green: 95% confidence interval).

### 6.2.6 Conclusions

- The time series was non-stationary, and stationarity was achieved through first differencing.
- The ARIMA(1,1,1) model was selected as the best model based on AIC.
- The model residuals do not exhibit significant autocorrelation and are approximately normally distributed.
- The forecast for the next 10 years shows a stabilized mean value but considerable uncertainty, as reflected by the wide confidence intervals.

**Overall, the ARIMA(1,1,1) model can be used for short- and medium-term forecasting of global land temperature, but forecast uncertainty must be treated with caution.**

## 6.3 Seasonality, Trend, and Cyclical Variation Analysis of Beer Sales (2016–2019)

The table presents the beer sales of an industrial company, in millions of bottles, by season from 2016 to 2019.

| Year | Winter | Spring | Summer | Autumn |
|---|---|---|---|---|
| 2016 | 1 | 3 | 6 | 4 |
| 2017 | 2 | 2 | 7 | 5 |
| 2018 | 2 | 4 | 8 | 5 |
| 2019 | 1 | 3 | 8 | 6 |

**Table 7:** Beer sales (millions of bottles) by season and year.

### 6.3.1 Seasonality Removal and Seasonal Indices

The data were first converted into a time series with frequency 4 (seasons per year), and decomposition was performed using a multiplicative model. The seasonal indices express how each season changes sales relative to the annual average.

The figure is not available.

**Figure 41:** Seasonal indices. Sales increase substantially in summer and decrease in winter.

*Interpretation:* The seasonal indices show that the highest sales occur in summer (index $> 1$), whereas the lowest sales occur in winter ($< 1$), which is expected for this product.

### 6.3.2 Trend Line

After deseasonalization, linear regression was applied to estimate the trend line and capture the overall direction of sales without the seasonal effect.

The figure is not available.

**Figure 42:** Sales trend line (green line) and deseasonalized sales (red line).

*Interpretation:* The trend line shows a small increase, indicating slow but steady growth in demand after seasonality is removed.

### 6.3.3 Cyclical Variation

The cyclical variation was estimated using relative cyclical residuals, defined as the ratio of deseasonalized sales to the trend line.

The figure is not available.

**Figure 43:** Cyclical variation plot (relative cyclical residuals).

*Interpretation:* The values fluctuate around 1, indicating that there are no strong cyclical effects, that is, no major additional fluctuations beyond seasonality and trend.

### 6.3.4 Raw Sales Time Series

The figure is not available.

**Figure 44:** Time series of the original raw sales before preprocessing.

*Conclusion:*

- There is strong seasonality, with a substantial increase in sales during summer.
- The trend line shows a slightly increasing sales trajectory over the years.
- The cyclical component does not show substantial deviations, so no strong cyclical effects are detected.

# 7 Overall Conclusions

This assignment analyzed five different real-world time series in order to understand their characteristics and reliably forecast future values using appropriate time series models.

The main conclusions are as follows:

- For the global land temperature data, the **ARIMA(1,1,1)** model provided the best fit according to the AIC/BIC criteria and diagnostic tests, while the forecast confidence intervals were relatively wide, reflecting the uncertainty of long-term forecasting.
- For the weekly mortality time series (*cmort*), the **AR(2)** model described the data structure satisfactorily, with residuals behaving as white noise.
- For monthly rainfall in Zevgolatio and the monthly unemployment rate in Spain, appropriate SARIMA models were applied, highlighting both seasonality and the long-term behavior of the data.
- In the beer-sales time series, the analysis of seasonality, trend, and cyclical variation revealed strong periodic fluctuations, with sales increasing substantially during summer.
- Overall, the time series analysis highlighted the importance of careful statistical investigation, including stationarity testing and residual diagnostics, as well as appropriate model selection in order to improve forecasts.

The results demonstrate that systematic analysis and suitable modeling can provide useful forecasts and support decision-making in a variety of domains, including climate, health, and the labor market.

# References


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# A Appendices

## A.1 Implementation Code

The following R code was used for the analysis of Spain’s unemployment time series:

**Listing 1:** R code for Spanish Unemployment Analysis

```
# 1.a
# Creation of a time series
unemployment <- c(13.6, 13.8, 13.8, 13.1, 12.5, 12.4, 12.4, 12.8, 13.0, 13.0, 13.0, 13.0,
                  13.6, 13.6, 13.0, 12.1, 11.6, 11.4, 11.7, 11.9, 12.1, 11.9, 11.7, 11.7,
                  12.3, 12.4, 12.2, 11.6, 11.2, 11.1, 11.2, 11.3, 11.1, 10.8, 10.5, 10.6,
                  11.2, 11.5, 11.5, 10.9)

# Creation of object time series
unemp_ts <- ts(unemployment, start = c(2022, 1), frequency = 12)

# 1. Timeline
plot(unemp_ts,
     main = "Unemployment Rate in Spain (Jan. 2022 - Apr. 2025)",
```

```r
     xlab = "Year",
     ylab = "Percentage (%)",
     col = "darkred",
     lwd = 2,
     ylim = c(10, 15))
grid(col = "gray", lty = "dotted")

# 2. Check of stationary series
# Calculation of first differences
diff_unemp <- diff(unemp_ts)

# Graph of first differences
plot(diff_unemp,
     main = "First Differences (Stagnation Index)",
     xlab = "Year",
     ylab = "Difference",
     col = "darkblue",
     lwd = 2)
grid(col = "gray", lty = "dotted")
abline(h = 0, col = "red", lty = 2)

# 3. Seasonality analysis
decomp <- decompose(unemp_ts)

# Graphical representation of decomposition
par(mfrow = c(4, 1), mar = c(3, 4, 2, 2))
plot(decomp$trend, main = "Trend", col = "darkgreen", lwd = 2)
plot(decomp$seasonal, main = "Seasonality", col = "darkblue", lwd = 2)
plot(decomp$random, main = "Random Element", col = "purple", lwd = 2)
plot(unemp_ts, main = "Real Series", col = "darkred", lwd = 2)
par(mfrow = c(1, 1))

# 4. Seasonality - boxplot by month
months <- factor(cycle(unemp_ts),
                 levels = 1:12,
                 labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"))

boxplot(unemp_ts ~ months,
        main = "Seasonal Unemployment Pattern (By Month)",
        xlab = "Month",
        ylab = "Unemployment Percentage (%)",
        col = "lightblue",
        border = "darkblue")
grid(nx = NA, ny = NULL, col = "gray", lty = "dotted")

# 5. Autocorrelation (ACF) for seasonality
```

```
acf(unemp_ts,
    main = "Autocorrelation Function (ACF)",
    lag.max = 36,
    col = "darkred",
    lwd = 2)
grid(col = "gray", lty = "dotted")

# 1.b
# First difference application
diff_unemp <- diff(unemp_ts)

# Graph of first differences
plot(diff_unemp,
     main = "First Differences of Unemployment (Stationary Series)",
     xlab = "Year",
     ylab = "Difference",
     col = "darkblue",
     lwd = 2)
abline(h = 0, col = "red", lty = 2)
grid(col = "gray", lty = "dotted")

# Autocorrelation Functions (ACF) and Partial Autocorrelation Functions (PACF)
par(mfrow = c(1, 2))

# ACF
acf(diff_unemp,
    lag.max = 24,
    main = "ACF: First Differences",
    col = "darkred",
    lwd = 2)

# PACF
pacf(diff_unemp,
     lag.max = 24,
     main = "PACF: First Differences",
     col = "darkgreen",
     lwd = 2)
par(mfrow = c(1, 1))

# Model fitting
arma21 <- arima(diff_unemp, order = c(2, 0, 1))  # ARMA(2,1) for the differences
arma11 <- arima(diff_unemp, order = c(1, 0, 1))  # ARMA(1,1) for the differences

# Summary of models
cat("Summary ARMA(2,1):\n")
print(summary(arma21))

```

```r
cat("\nSummary ARMA(1,1):\n")
print(summary(arma11))

# Balance check
check_residuals <- function(model) {
  par(mfrow = c(1, 2))
  acf(residuals(model), main = "ACF Residuals", col = "purple")
  pacf(residuals(model), main = "PACF Residuals", col = "orange")
  par(mfrow = c(1, 1))
}

cat("\nBalance Check ARMA(2,1):\n")
check_residuals(arma21)

cat("\nBalance Check ARMA(1,1):\n")
check_residuals(arma11)

# Best model
best_model <- arma21

# 6-month forecast
forecast_values <- predict(best_model, n.ahead = 6)

# Plot
plot(diff_unemp,
     main = "Forecast of First Differences (ARMA(2,1))",
     xlab = "Year",
     ylab = "Difference",
     xlim = c(2022, 2025.5),
     col = "darkblue",
     lwd = 2)
lines(ts(forecast_values$pred, start = end(diff_unemp), frequency =
    12),
      col = "red", lwd = 2)
legend("topright",
       legend = c("Actual", "Forecasts"),
       col = c("darkblue", "red"),
       lwd = 2)

#1.d
# --- FIT ARIMA(2,1,1) ---
arima_model <- arima(unemp_ts, order = c(2, 1, 1))

# --- RESULTS ANALYSIS ---
# Coefficients and p-values
coefs <- arima_model$coef
se <- sqrt(diag(arima_model$var.coef))
z_stats <- coefs / se
p_values <- 2 * (1 - pnorm(abs(z_stats)))
results <- data.frame(
  Estimate = coefs,
```

```
  Std.Error = se,
  z.value = z_stats,
  p.value = p_values
)
print(results)

# --- RESIDUAL DIAGNOSTICS ---
par(mfrow = c(1, 2))
acf(resid(arima_model), main = "ACF Residuals")
pacf(resid(arima_model), main = "PACF Residuals")
par(mfrow = c(1, 1))
# Ljung-Box test (12 lags)
box_test <- Box.test(resid(arima_model), type = "Ljung-Box", lag =
   12)
print(box_test)

# --- 12-MONTH FORECAST ---
forecast_values <- predict(arima_model, n.ahead = 12)

# --- PLOT ---
plot(unemp_ts,
     main = "Unemployment Forecast (ARIMA(2,1,1))",
     xlim = c(2022, 2026),
     ylim = c(10, 14),
     ylab = "Unemployment Rate (%)",
     xlab = "Year")
# Forecasts (red line)
lines(ts(forecast_values$pred, start = end(unemp_ts) + c(0, 1),
   frequency = 12), col = "red", lwd = 2)
# Confidence intervals (blue dashed lines)
lines(ts(forecast_values$pred + 1.96 * forecast_values$se, start =
   end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
lines(ts(forecast_values$pred - 1.96 * forecast_values$se, start =
   end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
# Legend
legend("topright",
       legend = c("Actual", "Forecast", "95% Interval"),
       col = c("black", "red", "blue"),
       lty = c(1, 1, 2),
       lwd = 2)

# --- Return to single plot ---
par(mfrow = c(1, 1))

#1.g

# --- Initial model (ARIMA(2,1,1)) ---
arima_211 <- arima(unemp_ts, order = c(2, 1, 1))
aic_211 <- AIC(arima_211)
lb_211 <- Box.test(resid(arima_211), type = "Ljung-Box", lag = 12)$
   p.value
```

```r

# --- Extended models ---
arima_311 <- arima(unemp_ts, order = c(3, 1, 1))
aic_311 <- AIC(arima_311)
lb_311 <- Box.test(resid(arima_311), type = "Ljung-Box", lag = 12)$p.value

arima_212 <- arima(unemp_ts, order = c(2, 1, 2))
aic_212 <- AIC(arima_212)
lb_212 <- Box.test(resid(arima_212), type = "Ljung-Box", lag = 12)$p.value

arima_221 <- arima(unemp_ts, order = c(2, 2, 1))
aic_221 <- AIC(arima_221)
lb_221 <- Box.test(resid(arima_221), type = "Ljung-Box", lag = 12)$p.value

# --- Summary table ---
models <- c("ARIMA(2,1,1)", "ARIMA(3,1,1)", "ARIMA(2,1,2)", "ARIMA(2,1,2)")
aic_values <- c(aic_211, aic_311, aic_212, aic_221)
lb_pvalues <- c(lb_211, lb_311, lb_212, lb_221)
summary_table <- data.frame(Model = models, AIC = aic_values, LjungBox_p = lb_pvalues)
print(summary_table)

# --- Residual check of the best model (lowest AIC) ---
best_model_index <- which.min(aic_values)
best_model_name <- models[best_model_index]
cat("Best model:", best_model_name, "\n")
if(best_model_index == 1) best_model <- arima_211
if(best_model_index == 2) best_model <- arima_311
if(best_model_index == 3) best_model <- arima_212
if(best_model_index == 4) best_model <- arima_221

# Residual plots
par(mfrow = c(1, 2))
acf(resid(best_model), main = paste("ACF Residuals", best_model_name))
pacf(resid(best_model), main = paste("PACF Residuals", best_model_name))
par(mfrow = c(1, 1))

#1.h

# 24-month forecast using the final model
forecast_values <- predict(arima_311, n.ahead = 24)

# Adjust the plot to your needs:
plot(unemp_ts,
     xlim = c(2022, 2027),
```

```r
     ylim = c(min(unemp_ts, forecast_values$pred - 1.96*forecast_values$se), max(unemp_ts, forecast_values$pred + 1.96*forecast_values$se)),
     main = "Unemployment Rate Forecast (24 Months Ahead)",
     ylab = "Unemployment Rate (%)",
     xlab = "Year")
lines(ts(forecast_values$pred, start = end(unemp_ts) + c(0, 1), frequency = 12), col = "red", lwd = 2)
lines(ts(forecast_values$pred + 1.96 * forecast_values$se, start = end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
lines(ts(forecast_values$pred - 1.96 * forecast_values$se, start = end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
legend("topright",
       legend = c("Historical", "Forecast", "95% Interval"),
       col = c("black", "red", "blue"),
       lty = c(1, 1, 2), lwd = 2)
```

**Listing 2:** R code for Exercise 2 - Monthly Rainfall Analysis with SARIMA

```r
#2.a

data <- read.csv("C:/Users/pathana/Documents/predictive analytics/open-meteo-37.93N22.83E37m.csv", skip = 2)
head(data)

names(data)
head(data)

data$date <- as.Date(substr(data$time, 1, 10))
data$year <- format(data$date, "%Y")
data$month <- format(data$date, "%m")

monthly_rain <- aggregate(data[,"rain..mm."],
                          by = list(year = data$year, month = data$month),
                          FUN = sum, na.rm = TRUE)
names(monthly_rain)[3] <- "rain_sum"

# Create time series
ts_rain <- ts(monthly_rain$rain_sum, start = c(as.numeric(monthly_rain$year[1]), as.numeric(monthly_rain$month[1])), frequency = 12)
plot(ts_rain, main = "Monthly Rainfall in Zevgolatio, Corinthia (2016 - 2025)", ylab = "mm", xlab = "Year")

#2.b
acf(ts_rain, main = "ACF of Monthly Rainfall")
pacf(ts_rain, main = "PACF of Monthly Rainfall")

# 2.c: First seasonal differences and stationarity analysis
seas_diff <- diff(ts_rain, lag = 12)
plot(seas_diff, main = "First Seasonal Differences (Lag=12)", ylab
```

```
= "Difference (mm)", xlab = "Year")
grid(col = "gray", lty = "dotted")

par(mfrow = c(1, 2))
acf(seas_diff, lag.max = 36, main = "ACF: Seasonal Differences", col = "darkblue")
pacf(seas_diff, lag.max = 36, main = "PACF: Seasonal Differences", col = "darkred")
par(mfrow = c(1, 1))

years <- unique(floor(time(seas_diff)))
means <- tapply(seas_diff, floor(time(seas_diff)), mean)
vars <- tapply(seas_diff, floor(time(seas_diff)), var)

cat("Annual Means:\n", round(means, 2), "\n")
cat("Annual Variances:\n", round(vars, 2), "\n")

if (max(vars)/min(vars) < 4 && diff(range(means)) < 2*sd(seas_diff)) {
  cat("Stationary Series (stable mean and variance)")
} else {
  cat("Non-Stationary Series")
}

#2.e
double_diff <- diff(diff(ts_rain, lag = 12), differences = 1)
plot(double_diff, main = "Double Differencing (First + Seasonal Differences)", ylab = "Difference (mm)", xlab = "Year", col = "darkblue", lwd = 2)
grid(col = "gray", lty = "dotted")

par(mfrow = c(1, 2))
acf(double_diff, lag.max = 36, main = "ACF: Double Differences", col = "darkred")
pacf(double_diff, lag.max = 36, main = "PACF: Double Differences", col = "darkgreen")
par(mfrow = c(1, 1))

#2.f
fit_sarima <- stats::arima(
  ts_rain,
  order = c(2, 1, 1),
  seasonal = list(order = c(1, 1, 1), period = 12)
)

cat("Model Parameters:\n")
print(fit_sarima$coef)
cat("\nStatistics:\n")
cat("AIC:", fit_sarima$aic, "\n")
cat("log-likelihood:", fit_sarima$loglik, "\n")

```

```r
residuals_sarima <- residuals(fit_sarima)
acf_plot <- function(res, main = "ACF Residuals") {
  n <- length(res)
  acf_vals <- acf(res, plot = FALSE)$acf[-1]
  lags <- 1:length(acf_vals)
  plot(lags, acf_vals, type = "h", main = main, ylim = c(-1, 1),
     xlab = "Lag", ylab = "ACF")
  abline(h = 0, col = "black")
  abline(h = c(-1.96, 1.96)/sqrt(n), col = "blue", lty = 2)
}
pacf_plot <- function(res, main = "PACF Residuals") {
  n <- length(res)
  pacf_vals <- pacf(res, plot = FALSE)$acf
  lags <- 1:length(pacf_vals)
  plot(lags, pacf_vals, type = "h", main = main, ylim = c(-1, 1),
     xlab = "Lag", ylab = "PACF")
  abline(h = 0, col = "black")
  abline(h = c(-1.96, 1.96)/sqrt(n), col = "blue", lty = 2)
}
par(mfrow = c(1, 3))
acf_plot(residuals_sarima)
pacf_plot(residuals_sarima)
hist(residuals_sarima, main = "Residual Histogram", col = "
   lightblue")
par(mfrow = c(1, 1))

ljung_box_test <- function(res, lag = 24) {
  n <- length(res)
  lb_test <- Box.test(res, lag = lag, type = "Ljung-Box")
  cat("Ljung-Box test (lag =", lag, "):\n")
  cat("X-squared =", lb_test$statistic, " p-value =", lb_test$p.
     value, "\n")
  if (lb_test$p.value > 0.05) {
    cat("No autocorrelation (good residuals)\n")
  } else {
    cat("Autocorrelation is present (problematic residuals)\n")
  }
}
ljung_box_test(residuals_sarima)

future_forecast <- predict(fit_sarima, n.ahead = 12)
last_obs_time <- tail(time(ts_rain), 1)
forecast_dates <- last_obs_time + (1:12)/12
plot(ts_rain, main = "Monthly Rainfall Forecast", xlab = "Year",
   ylab = "Rainfall (mm)", xlim = c(start(ts_rain)[1], last_obs_
   time + 13/12), ylim = c(0, max(c(ts_rain, future_forecast$pred +
    1.96*future_forecast$se), na.rm = TRUE)))
last_obs_value <- tail(ts_rain, 1)
lines(c(last_obs_time, forecast_dates[1]), c(last_obs_value, future
   _forecast$pred[1]), col = "red", lwd = 2)
lines(forecast_dates, future_forecast$pred, col = "red", lwd = 2)
```

```
lines(forecast_dates, future_forecast$pred + 1.96 * future_forecast
   $se, col = "blue", lty = 2)
lines(forecast_dates, future_forecast$pred - 1.96 * future_forecast
   $se, col = "blue", lty = 2)
legend("topleft",
       legend = c("Observations", "Forecast", "95% Confidence
          Interval"),
       col = c("black", "red", "blue"),
       lty = c(1, 1, 2),
       cex = 0.8)

#2.g
residuals_sarima <- residuals(fit_sarima)
par(mfrow = c(2, 2))
plot(residuals_sarima, main = "Residuals over Time", ylab = "
   Residuals")
abline(h = 0, col = "red")
acf(residuals_sarima, main = "ACF Residuals", col = "darkblue")
pacf(residuals_sarima, main = "PACF Residuals", col = "darkred")
qqnorm(residuals_sarima, main = "Q-Q Plot")
qqline(residuals_sarima, col = "blue")
par(mfrow = c(1, 1))
Box.test(residuals_sarima, lag = 24, type = "Ljung-Box")

original_model <- fit_sarima
original_aic <- original_model$aic
model_ar_up <- arima(ts_rain, order = c(3, 1, 1), seasonal = list(
   order = c(1, 1, 1), period = 12))
model_ma_up <- arima(ts_rain, order = c(2, 1, 2), seasonal = list(
   order = c(1, 1, 1), period = 12))
model_sar_up <- arima(ts_rain, order = c(2, 1, 1), seasonal = list(
   order = c(2, 1, 1), period = 12))
model_sma_up <- arima(ts_rain, order = c(2, 1, 1), seasonal = list(
   order = c(1, 1, 2), period = 12))
model_full_up <- arima(ts_rain, order = c(3, 1, 2), seasonal = list
   (order = c(2, 1, 2), period = 12))

models <- list(
  "Original" = original_model,
  "AR+1" = model_ar_up,
  "MA+1" = model_ma_up,
  "SAR+1" = model_sar_up,
  "SMA+1" = model_sma_up,
  "Full+1" = model_full_up
)
compare_models <- function(model_list) {
  results <- data.frame(
    Model = names(model_list),
    AIC = sapply(model_list, function(m) m$aic),
    LogLik = sapply(model_list, function(m) m$loglik),
    stringsAsFactors = FALSE
```

```
  )
  results$DeltaAIC <- results$AIC - original_aic
  results[order(results$AIC), ]
}
model_comparison <- compare_models(models)
print(model_comparison)
check_residuals <- function(model) {
  res <- residuals(model)
  par(mfrow = c(1, 3))
  acf(res, main = "ACF Residuals", col = "darkblue")
  pacf(res, main = "PACF Residuals", col = "darkred")
  qqnorm(res, main = "Q-Q Plot")
  qqline(res, col = "blue")
  par(mfrow = c(1, 1))
  lb_test <- Box.test(res, lag = 24, type = "Ljung-Box")
  cat("Ljung-Box test p-value:", lb_test$p.value, "\n")
}
best_model_name <- model_comparison$Model[which.min(model_comparison$AIC)]
best_model <- models[[best_model_name]]
cat("\nResidual Check for the Best Model:", best_model_name, "\n")
check_residuals(best_model)
if (model_comparison$DeltaAIC[model_comparison$Model == best_model_name] < -2) {
  cat("\nThe initial model is NOT adequate. The best model is", best_model_name)
} else {
  cat("\nThe initial model is adequate.")
}

#2.h
forecast_24m <- predict(fit_sarima, n.ahead = 24)
last_obs_time <- end(ts_rain)
last_obs_value <- tail(ts_rain, 1)
forecast_times <- seq(last_obs_time[1] + (last_obs_time[2])/12 + 1/12, by = 1/12, length.out = 24)
plot(ts_rain,
     main = "Monthly Rainfall Forecast (2 Years)",
     xlab = "Year",
     ylab = "Rainfall (mm)",
     xlim = c(start(ts_rain)[1], last_obs_time[1] + 3),
     ylim = c(0, max(ts_rain, forecast_24m$pred + 1.96*forecast_24m$se, na.rm = TRUE)),
     lwd = 2)
segments(x0 = last_obs_time[1] + (last_obs_time[2]-1)/12, y0 = last_obs_value,
         x1 = forecast_times[1], y1 = forecast_24m$pred[1], col = "red", lwd = 2)
lines(forecast_times, forecast_24m$pred, col = "red", lwd = 2)
lines(forecast_times, forecast_24m$pred + 1.96 * forecast_24m$se, col = "blue", lty = 2)
```

```
lines(forecast_times, forecast_24m$pred - 1.96 * forecast_24m$se, col = "blue", lty = 2)
legend("topleft",
       legend = c("Historical Data", "SARIMA Forecast", "95% Confidence Interval"),
       col = c("black", "red", "blue"),
       lty = c(1, 1, 2),
       lwd = c(2, 2, 1),
       cex = 0.8)
abline(v = forecast_times[1], col = "green", lty = 2)
text(x = forecast_times[1], y = max(ts_rain)*0.9, "Forecast Start", pos = 4, col = "darkgreen", cex = 0.8)

year1 <- forecast_24m$pred[1:12]
year2 <- forecast_24m$pred[13:24]
historical_seasonal <- aggregate(as.vector(ts_rain) ~ cycle(ts_rain), FUN = mean)
par(mfrow = c(1, 2))
plot(1:12, year1, type = "o", col = "red", main = "Forecast Year 1 vs Historical", xlab = "Month", ylab = "Rainfall (mm)", ylim = range(historical_seasonal[,2], year1))
lines(1:12, historical_seasonal[,2], type = "o", col = "blue")
legend("topright", legend = c("Forecast", "Historical"), col = c("red", "blue"), lty = 1)
plot(1:12, year2, type = "o", col = "red", main = "Forecast Year 2 vs Historical", xlab = "Month", ylab = "Rainfall (mm)", ylim = range(historical_seasonal[,2], year2))
lines(1:12, historical_seasonal[,2], type = "o", col = "blue")
par(mfrow = c(1, 1))
yearly_avg <- c(mean(year1), mean(year2))
se_avg <- c(mean(forecast_24m$se[1:12]), mean(forecast_24m$se[13:24]))
plot(1:2, yearly_avg, type = "b", col = "red", main = "Annual Forecast Mean", xlab = "Forecast Year", ylab = "Mean Rainfall (mm)", ylim = c(min(yearly_avg - 1.96*se_avg), max(yearly_avg + 1.96*se_avg)))
arrows(x0 = 1:2, y0 = yearly_avg - 1.96*se_avg, x1 = 1:2, y1 = yearly_avg + 1.96*se_avg, angle = 90, code = 3, length = 0.1)
abline(h = mean(ts_rain), col = "blue", lty = 2)
legend("topright", legend = c("Forecast", "Historical Mean"), col = c("red", "blue"), lty = c(1, 2))

diff_year1 <- year1 - historical_seasonal[,2]
diff_year2 <- year2 - historical_seasonal[,2]
t_test_year1 <- t.test(diff_year1)
t_test_year2 <- t.test(diff_year2)
res_acf <- acf(forecast_24m$pred, main = "ACF of Forecasts")
pred_range <- apply(rbind(forecast_24m$pred - 1.96*forecast_24m$se, forecast_24m$pred + 1.96*forecast_24m$se), 2, diff)
cat("Difference Year 1:", round((mean(year1) - mean(ts_rain))/mean(ts_rain)*100, 1), "%\n")
```

```
cat("Difference Year 2:", round((mean(year2) - mean(ts_rain))/mean(
   ts_rain)*100, 1), "%")

test_size <- 24
train <- window(ts_rain, end = c(end(ts_rain)[1]-2, 12))
test <- window(ts_rain, start = c(end(ts_rain)[1]-1, 1))
fit_train <- arima(train, order = c(2,1,1), seasonal = list(order =
    c(1,1,1), period = 12))
test_forecast <- predict(fit_train, n.ahead = test_size)
errors <- test - test_forecast$pred
mae <- mean(abs(errors))
rmse <- sqrt(mean(errors^2))
plot(test, main = "Actual vs Predicted Values")
lines(ts(test_forecast$pred, start = start(test), frequency = 12),
   col = "red")
```

**Listing 3:** Exercise 3 - Time Series Analysis with AR, ARIMA and Seasonal Decomposition

```
# 3.1 Mortality forecasting with AR(2)

library(astsa)

# Create the time series
ts_cmort <- ts(cmort, frequency = 52)

# Fit AR(2)
ar2_fit <- arima(ts_cmort, order = c(2,0,0))

# 4-week forecast
forecast_horizon <- 4
forecast <- predict(ar2_fit, n.ahead = forecast_horizon)

# 95% confidence intervals
upper_bound <- forecast$pred + 1.96 * forecast$se
lower_bound <- forecast$pred - 1.96 * forecast$se

# Plot results
n <- length(ts_cmort)
weeks <- (n - 30):(n + forecast_horizon)
plot(weeks[1:31], ts_cmort[(n-30):n], type = "l", lwd = 2,
     ylim = range(ts_cmort[(n-30):n], upper_bound, lower_bound),
     xlim = range(weeks),
     main = "AR(2) Mortality Forecast",
     xlab = "Week", ylab = "Mortality")
forecast_weeks <- (n + 1):(n + forecast_horizon)
lines(forecast_weeks, forecast$pred, col = "red", lwd = 2, type = "
   b", pch = 16)
lines(forecast_weeks, upper_bound, col = "blue", lty = 2, lwd = 2)
lines(forecast_weeks, lower_bound, col = "blue", lty = 2, lwd = 2)
legend("topright",
       legend = c("Historical", "AR(2) Forecast", "95% Confidence
```

```
          Interval"),
       col = c("black", "red", "blue"),
       lty = c(1, 1, 2), pch = c(NA, 16, NA), lwd = c(2,2,2), bty =
          "n")

# Residual diagnostics
resid_ar2 <- residuals(ar2_fit)
par(mfrow = c(2,3))
plot(resid_ar2, type = "l", main = "AR(2) Residuals", ylab = "
   Residuals")
abline(h = 0, col = "red", lty = 2)
acf(resid_ar2, main = "ACF Residuals")
pacf(resid_ar2, main = "PACF Residuals")
hist(resid_ar2, breaks = 20, col = "lightblue", main = "Residual
   Histogram")
qqnorm(resid_ar2); qqline(resid_ar2, col = "red")
plot(weeks[1:31], ts_cmort[(n-30):n], type = "l", lwd = 2, main = "
   Latest Values", xlab = "Week", ylab = "Mortality")
par(mfrow = c(1,1))

# Ljung-Box test
lb <- Box.test(resid_ar2, lag = 10, type = "Ljung-Box")
cat("Ljung-Box test p-value:", lb$p.value, "\n")
cat("Coefficients:\n"); print(ar2_fit$coef)
cat("AIC:", ar2_fit$aic, "\n")


# 3.2 Global temperature analysis and forecasting with ARIMA

library(astsa)
data(gtemp_land)
years <- 1850:2023
x_ts <- ts(gtemp_land, start = 1850, frequency = 1)

# Original plot
plot(x_ts, main = "Global Land Temperature (1850-2023)", ylab = "
   Anomaly (degC)", xlab = "Year")

# ACF/PACF of the original series and first difference
par(mfrow = c(1,2))
acf(x_ts, main = "ACF of Original Series")
pacf(x_ts, main = "PACF of Original Series")
par(mfrow = c(1,1))
dx_ts <- diff(x_ts)
plot(dx_ts, main = "First Difference of Series", ylab = "Difference
    from Previous Year", xlab = "Year")
par(mfrow = c(1,2))
acf(dx_ts, main = "ACF of First Difference")
pacf(dx_ts, main = "PACF of First Difference")
par(mfrow = c(1,1))

```

```
# ARIMA(1,1,1) model
fit <- arima(x_ts, order = c(1,1,1))
resid_fit <- residuals(fit)

# Residual diagnostics
par(mfrow = c(2,2))
plot(resid_fit, main = "ARIMA Residuals", ylab = "Residuals")
acf(resid_fit, main = "ACF of Residuals")
pacf(resid_fit, main = "PACF of Residuals")
hist(resid_fit, main = "Residual Histogram", col = "lightblue")
par(mfrow = c(1,1))

# Ljung-Box test
lb_test <- Box.test(resid_fit, lag = 20, type = "Ljung-Box")
cat("Ljung-Box p-value:", lb_test$p.value, "\n")
cat("AIC:", fit$aic, "\n")
cat("Coefficients:\n")
print(fit$coef)

# Forecast for 10 years
n_forecast <- 10
pred <- predict(fit, n.ahead = n_forecast)
z <- qnorm(0.975)
upper <- pred$pred + z * pred$se
lower <- pred$pred - z * pred$se
last_year <- end(x_ts)[1]
forecast_years <- (last_year + 1):(last_year + n_forecast)
forecast_years <- c(last_year, forecast_years)
forecast_pred <- c(tail(x_ts, 1), pred$pred)
upper <- c(tail(x_ts, 1), upper)
lower <- c(tail(x_ts, 1), lower)

# Visualization
plot(window(x_ts, start = 2000), type = "o", pch = 16, col = "blue",
     xlim = c(2000, last_year + n_forecast),
     ylim = range(c(x_ts, lower, upper), na.rm = TRUE),
     main = "Global Land Temperature Forecast",
     xlab = "Year", ylab = "Anomaly (degC)")
lines(forecast_years, forecast_pred, type = "o", col = "red", pch = 18, lty = 2)
lines(forecast_years, upper, col = "darkgreen", lty = 3)
lines(forecast_years, lower, col = "darkgreen", lty = 3)
polygon(c(forecast_years, rev(forecast_years)), c(upper, rev(lower)),
        col = rgb(0, 0.5, 0, 0.2), border = NA)
legend("topleft", legend = c("Historical", "Forecast", "95% CI"),
       col = c("blue", "red", "darkgreen"),
       lty = c(1, 2, 1), pch = c(16, 18, NA), bg = "white")
```

```r
# 3.3 Seasonal analysis and trend of water sales

# Enter data
sales_data <- data.frame(
  Year = c(2016, 2017, 2018, 2019),
  Winter = c(1, 2, 2, 1),
  Spring = c(3, 2, 4, 3),
  Summer = c(6, 7, 8, 8),
  Autumn = c(4, 5, 5, 6)
)

# Convert to time series
sales_ts <- ts(c(t(sales_data[, -1])), start = 2016, frequency = 4)

# Calculate seasonal indices
decompose_sales <- decompose(sales_ts, type = "multiplicative")
seasonal_indices <- decompose_sales$seasonal[1:4]
cat("Seasonal Indices:\n")
print(round(seasonal_indices, 3))

# Deseasonalization
deseasonalized_sales <- sales_ts / seasonal_indices

# Trend line
time_points <- time(sales_ts)
trend_model <- lm(deseasonalized_sales ~ time_points)
trend_line <- ts(fitted(trend_model), start = 2016, frequency = 4)

# Cyclical component
cyclic_component <- deseasonalized_sales / trend_line

# Visualization
par(mfrow = c(2,2))
plot(sales_ts, main = "Raw Sales", ylab = "Sales (million bottles)", col = "blue")
legend("topleft", "Raw", col = "blue", lty = 1)
plot(deseasonalized_sales, main = "Deseasonalized Sales", ylab = "Sales", col = "red")
lines(trend_line, col = "darkgreen", lwd = 2)
legend("topleft", c("Deseasonalized", "Trend"), col = c("red", "darkgreen"), lty = c(1,1))
plot(cyclic_component, main = "Cyclical Variation", ylab = "Relative Residuals", col = "purple")
abline(h = 1, col = "gray", lty = 2)
legend("topleft", "Cyclical", col = "purple", lty = 1)
barplot(seasonal_indices, names.arg = c("Winter", "Spring", "Summer", "Autumn"),
        main = "Seasonal Indices", ylab = "Index")
par(mfrow = c(1,1))
```